THE UNIVERSITY OF ZAMBIA
Department of Mathematics and Statistics
MAT1110: Foundation Mathematics and Statistics for Social Sciences
Tutorial Sheet 1 (2021/2022)

1. (a) List the elements of the following sets

   i. $A = \{x : x \text{ is even}, x < 18, x \in \mathbb{N}\}$.
   ii. $B = \{x : -2 \leq x \leq \frac{7}{2}, x \in \mathbb{Z}\}$
   iii. $C = \{x : 4 + x = 3, x \in \mathbb{N}\}$
   iv. $D = \{x : x \text{ is a multiple of } 3, x < 10\}$.
   v. $F = \{x : x^2 = 4,\ x \in \mathbb{Z}.\}$

   (b) Let the universal set $U = \{x : x < 10,\ x \in \mathbb{N}\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$, $C = \{5, 6, 7, 8, 9\}$, $D = \{x : x = 2k + 1,\ k \in \mathbb{N}\}$, $E = \{x : x = 2k,\ k \in \mathbb{N}\}$, and $F = \{1, 5, 9\}$.

   i. List down the sets U, D and E, given that D and E are subsets of U.
   Hence, find
   ii. $A \cap (C \cup E)$.
   iii. $B - A$
   iv. $(A \cap B) - B$
   v. $F - (A \cap D)^c$

   (c) Let $U = \{x : x \in \mathbb{N}, 1 \leq x \leq 21\}$. Given that $A = \{x : x = 2k + 1, x < 20, k \in \mathbb{N}\}$ and $B = \{1, 3, 8, 19, 20\}$ and $C = \{1, 2, 6, 8, 9, 11, 15\}$.
   Illustrate the information on a Venn diagram. Hence, find the following;

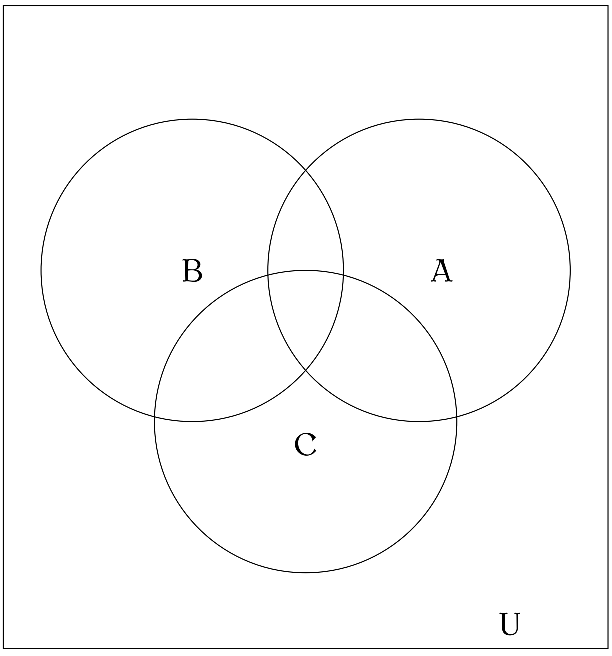
   i. $A^c \cap B^c$
   ii. $(A \cup B) \cap C$
   iii. $A - (B \cup C)$
   iv. $(A \cup B)^c$

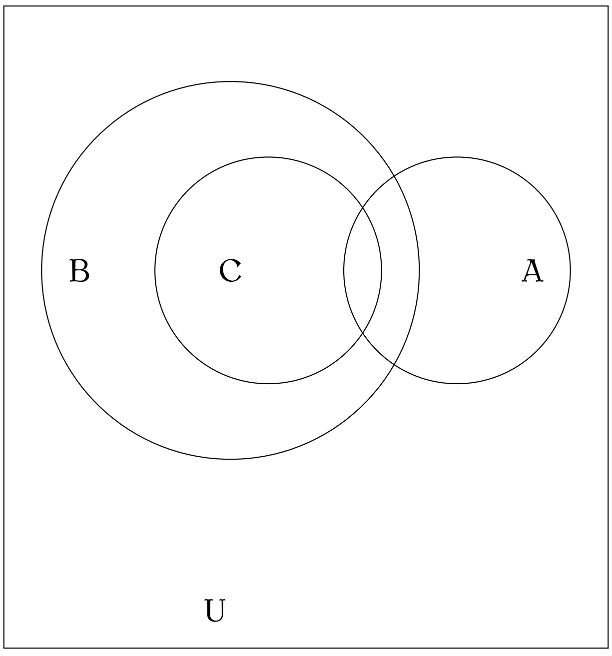
   (d) List down all the subsets of each of the following.

   i. $A = \{a\}$
   ii. $B = \{a, b\}$
   iii. $C = \{a, b, c\}$

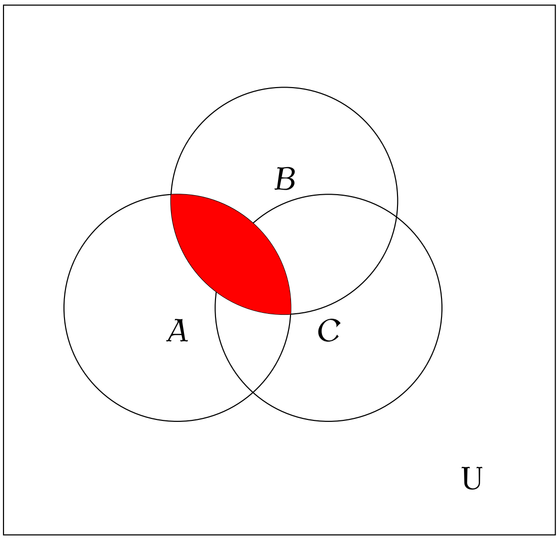
2. (a) In each of the following Venn diagrams, shade

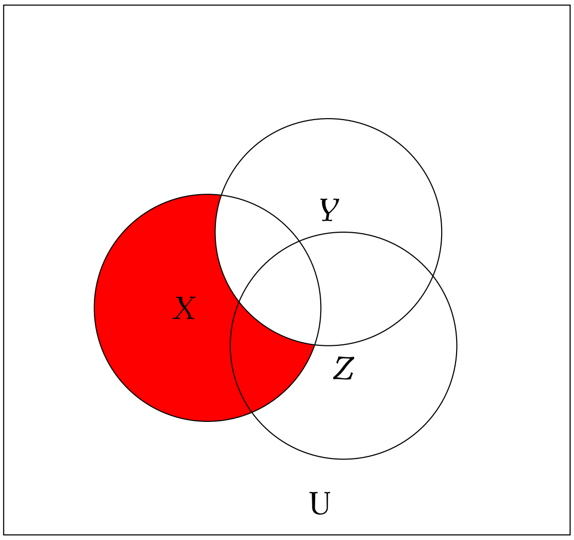
   i. $B - C$.
   ii. $A \cap (B \cap C^c)^c$.
   iii. $C \cap (A - B)^c$.





   (b) In each of the following Venn diagrams, use the set operations, $\cup$ and $\cap$ to describe the shaded region/s.





3. (a) If $A \subset B$, then simplify each of the following:

   i. $A \cap B^c$

   ii. $(B \cup A^c) \cap A$.

   (b) If A and B are disjoint, simplify each of the following:

   i. $A \cap B$.

   ii. $A \cup B$.

   iii. $(A \cap B)^c$

   iv. $A^c \cup B^c$.

   (c) Express each of the following in its simplest form:

   i. $(A \cup B) \cap (A \cup B^c)$

   ii. $[(X \cap Y)^c \cup (X - Y)]^c$.

   iii. $B^c \cap (A \cup B)$.

   iv. $[A^c \cup (A \cap B^c)]^c$.

4. (a) Let $\mathbb{R}$ be the universal set. Rewrite each of the following in set builder notation

   i. $A = (-5, 3)$

   ii. $B = (4, 7]$

   iii. $C = [0, 4]$

   iv. $D = [-2, 6)$

   (b) Let $\mathbb{R}$ be the universal set and $A = (-7, 3]$, $B = (0, 8)$, $C = \{x : x \leq 10,\ x \in \mathbb{R}\}$. Find each of the following sets and hence illustrate the answers on the number line.

   i. $A \cap C^c$.

   ii. $B - A$.

   iii. $A \cap (B \cup C^c)$.

   iv. $[A^c - (B - C)]^c$

   v. $A^c \cap B^c$.

   (c) Let $A = (-7, 3]$, $B = (0, 8)$. Illustrate De Morgan's laws using sets A and B.